



Aura by Centro

Enterprise AI Co-Pilot · CTO Briefing · Proof of Concept

Helping businesses drive meaningful change for growth.

STATUS: WORKING PROTOTYPE

WHAT IT IS

A multi-tenant AI co-pilot for Centro's BPO operations. Runs a local Gemma LLM, streams over WebSockets, and integrates with Zoho / Odoo / Genesys via the Model Context Protocol. Dual RAG + CAG memory model.

STACK & DEPLOYMENT

Frontend Next.js + Tailwind -> AWS (Amplify / CloudFront)
Backend FastAPI (async) -> AWS ECS / App Runner
LLM Local Gemma (vLLM) -> AWS EC2 GPU
Vectors Qdrant (self-host) -> zero licensing cost
Clients Web · branded desktop app · embedded in Zoho People

WHY IT MATTERS

Cost \$0 software licensing; only AWS GPU compute.
Privacy Data never leaves Centro; tenant isolation in-database.
Safety Every write requires explicit human confirmation.
Agility Knowledge & integrations update via files, not code.

1 Metadata-Enforced Sandboxing

RBAC scope compiled into a hard filter inside the vector DB. A Coastline query is physically unable to match Trueblue data.

2 Semantic CAG Cache

Cosine match ≥ 0.92 answers instantly and bypasses the LLM — sub-second, zero GPU cost for common questions.

3 Dual-Confirmation Action Cards

Mutations (payroll, scheduling, approvals) render a card showing target system, payload & risk. Nothing fires without a confirm.

4 Dynamic Schema Retrieval

Integration contracts live as files / WorkDrive, not in prompts. Endpoints change with no code change and no redeploy.